A GraphCast-style fitting model as Plexus2 operators

`prototype/graphcast` — nothing promoted

1. Mechanism

Every model in `connectome-gnn` is a one-step message-passing network on a *given* connectome. Two things wanted now sit outside that: a deeper, edge-stateful processor (GraphCast's), and datasets with no connectome at all. ZAPBench has 71,721 neurons of which about 481 are matched to an electron-microscopy reconstruction; the redox organoid is a three-dimensional field with no segmentation. In both, the coupling has to come from *space* rather than from wiring.

The arithmetic that decides what can be represented: a ZAPBench volume is acquired plane by plane, 72 planes at 12.0 ms, so one volume takes 0.851 s against a 0.914 s frame interval — 93% of the frame period. Neurons at different depths are therefore *not* sampled simultaneously, which is precisely what a message-passing operator assumes. That fact is data, not a modelling choice: the per-plane acquisition times are recorded, and the model is conditioned on the offset rather than pretending it away.

2. Equations and symbols

To be written at stage 2, when the operator exists. The two message forms this prototype selects between are

$$\texttt{simple} \quad \mathbf{m}_i = \sum_j W_{ij}\, g_\phi(v_j, a_j), \qquad \dot{v}_i = f_\theta(v_i, a_i, \mathbf{m}_i, b_i s(t)) \tag{1}$$

$$\texttt{graphcast} \quad e_{ij} \leftarrow e_{ij} + \mathrm{LN}\big(\mathrm{MLP}_e([e_{ij} \| v_j \| v_i])\big), \qquad v_i \leftarrow v_i + \mathrm{LN}\big(\mathrm{MLP}_v([v_i \| \textstyle\sum_j e_{ij}])\big) \tag{2}$$

with a_i the per-neuron embedding carrying heterogeneity invisible in the state, and b_i the per-neuron gain on the external stimulus $s(t)$.

3. The Plexus2 container

To be written as the operators land. Sets: `neuron`. Fields: `stim` (playback), `mesh` (only when the encoder/decoder option is on). The four architectural choices are *config options*, not forks — `encoder_decoder`, `message`, `n_passes`, `embedding` — which is why the gate table below counts 24 combinations.

4. Gates

A gate is a number with a threshold decided *before* the run and a stated consequence if it fails. A threshold chosen after seeing the number is not a threshold, so every threshold below is a literal in `gates.py` and predates the code it gates. Three tiers, and only the third is a statement about biology; this table is 9 bookkeeping / 8 closed form / 4 measurement, so two thirds of it is verification. That is the expected shape and is stated rather than hidden.

A gate's threshold belongs in the unit of the phenomenon, not of the mesh — a threshold in grid cells is denominated in the currency in which it is easiest to pass, and a green row of that kind reads as an endorsement the interface has not earned. Every row therefore carries what its number is *of*, and the measurement tier is only available because the spec declares `general.units:`.

	gate	threshold	measured	outcome
bookkeeping — does the code do what the operator says?				
G1	all four options PARSE (24 combinations)	24 of 24 option combinations load	24	**PASS**
G1b	all four options BUILD and take one step	24 of 24 option combinations run one forward step	—	SKIP
G2	nothing dataset-specific is hardcoded	0 offending literals outside config/	0	**PASS**
G3	scatter->gather round trip on a constant field	< 1e-6 of the field value	—	SKIP
G4	transfer weights are a partition of unity	|sum(w) - 1| < 1e-6	—	SKIP
G5	simple + 1 pass + no enc/dec is arithmetically NeuralGNN	< 1e-5 of the voltage range	—	SKIP
G6	residual blocks start at identity: 1 vs 16 passes at init	bit-identical (max |delta| == 0)	—	SKIP
G7	units declared, and every measurement threshold is in phenomenon units	1 = declared and checked	1	**PASS**
G8	a K=20 rollout on the toy does not diverge	state norm stays < 2x the ground-truth norm	—	SKIP
closed form — does it reproduce the physics it was given?				
G9	recover the spatial interaction kernel (toy_small)	R^2 > 0.90 against the GT kernel	—	SKIP
G10	recover per-type time constants, GT graph supplied	R^2 > 0.95 against the known tau	—	SKIP
G11	a_i recovers the types (toy_small)	ARI > 0.70 against the 8 true types	—	SKIP
G12	a_i recovers the types (toy_large)	ARI > 0.70 against the 65 true types	—	SKIP
G13	recover the per-neuron stimulus gain b_i	R^2 > 0.90 against the true gain	—	SKIP
G14	encoder/decoder is a genuine option: on vs off	|delta R^2(kernel)| < 0.03	—	SKIP
G15	graphcast vs simple message is RESOLVED, either way	|delta| reported against the 3-seed floor; below it is UNRESOLVED, not ranked	—	SKIP
G16	types are spatially mixed by construction	spatial-cell type purity within 20% of chance (1/n_types)	—	SKIP
measurement — does it agree with something observed?				
G17	ZAPBench held-out prediction of d(dF/F)/dt	R^2 > 0.268, the parameter-free kNN spatial pool	—	SKIP
G18	the learned stimulus gain b_i is spatially structured	Moran's I > 0.2 over the soma graph, against a permutation null	—	SKIP
G19	fitted calcium decay time constant	0.5 - 2 s (GCaMP6)	—	SKIP
G20	redox field fit reproduces the washout response	THRESHOLD TO BE FIXED from Development_Time_Trend.xlsx, before the run	—	SKIP

What the gates license, so far. Only stage 0 has run. G1 establishes that all 24 option combinations parse, G2 that no dataset path, name or dimension appears anywhere in the prototype's code, and G7 that the spec declares its units — so the measurement-tier rows below are *available* to be denominated in seconds and micrometres rather than in cells. Nothing about any model is licensed yet: no operator has been built, and every closed-form and measurement row is still SKIP.

G5 is the one to watch, because it is what will establish that the `simple` option is arithmetically the existing `NeuralGNN` and therefore that everything after it is a controlled variation on something already known to work.